# Multi-Modal Machine Learning Approach for Threat Detection in WhatsApp and Telegram: Spam Links, Document Based Malware, Chat Behavior, and Sensitive Content Analysis

ML Based sensitive content detection and alert system for WhatsApp and Telegram

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Project Proposal Report

B.Sc. (Hons) in Information Technology Specializing in Cyber Security

Sri Lanka Institute of Information Technology Sri Lanka

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## Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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## Abstract

The rapid growth of messaging platforms like WhatsApp and Telegram has increased the risk of users—especially minors—being exposed to inappropriate or sensitive multimedia content. To address this challenge, this component of the project focuses on developing a lightweight, machine learning (ML)–based detection and alert system for images and videos shared through these platforms. The proposed solution will be integrated into a cross-platform mobile application, functioning as a parental control–like feature.

The system will scan media files before they are sent. When a user selects an image, it will automatically undergo sensitive content detection; for videos, playback and scanning will occur before transmission. Instead of disabling or enabling the send button, the system will employ a floating UI element that overlays on top of the messaging interface. This floating UI will display the detection results—such as sensitivity alert—without interrupting the natural workflow. Users will be able to interact with the floating UI to acknowledge alerts and decide whether to proceed with sending.

The detection engine will utilize optimized deep learning models trained on curated datasets of sensitive and non-sensitive media. It will be designed for mobile performance, ensuring real-time scanning without compromising user experience. The floating UI will provide privacy-aware feedback through tiered alerts such as warnings, soft blocks, or permissions, depending on the confidence level of the detection.

This research contributes by introducing an open-source framework for sensitive content detection in encrypted messaging platforms, an area often dominated by closed commercial tools. The solution emphasizes ethical handling of media, lightweight deployment, and community-driven improvement to enhance online safety.

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## List of Abbreviations

| | | |
|---|---|---|
| ML | - | Machine Learning |
| AI | - | Artificial Intelligence |
| NLP | - | Natural Language Processing |
| CV | - | Computer Vision |
| API | - | Application Programming Interface |
| UI | - | User Interface |
| UX | - | User Experience |
| TP | - | True Positive |
| FP | - | False Positive |
| FN | - | False Negative |
| TN | - | True Negative |
| IoT | - | Internet of Things (if needed for integration context) |
| GPU | - | Graphics Processing Unit |
| CPU | - | Central Processing Unit |
| JSON | - | JavaScript Object Notation (common for data exchange in mobile apps) |
| SDK | - | Software Development Kit |
| OCR | - | Optical Character Recognition (if text detection is included) |
| FPS | - | Frames Per Second (video scanning performance) |
| F1-score | - | Harmonic mean of precision and recall (model evaluation metric) |

# I. INTRODUCTION

## 1. Background and Literature Review

The popularity of instant messaging platforms such as WhatsApp and Telegram has transformed communication, particularly among younger users. However, this convenience also exposes users to sensitive and inappropriate content, including nudity, violence, and explicit media, which can negatively affect emotional well-being and lead to digital exploitation. Studies have shown that minors and inexperienced users are particularly vulnerable because of their lack of awareness and the absence of effective parental monitoring tools on these platforms.

Current messaging applications provide limited controls for filtering multimedia content. While some commercial solutions exist, most are closed-source, expensive, or not designed for real-time integration with encrypted platforms. Additionally, many parental control apps operate at a device or network level, often requiring intrusive permissions, which can affect usability and privacy.

Recent advancements in machine learning (ML) and computer vision (CV) provide opportunities to automatically analyze multimedia content before transmission. Pre-trained deep learning models such as convolutional neural networks (CNNs), MobileNet, and EfficientNet have demonstrated high accuracy in image and video classification tasks, including sensitive content detection. Mobile-optimized frameworks now enable lightweight deployment, making real-time analysis on smartphones feasible without significant performance degradation.

Research in this domain focuses on three main areas:

- Sensitive media detection using ML – Models trained on curated datasets (e.g., NSFW image detection, violent content classification).

- Privacy-aware parental controls – Tools that analyze media without storing or sharing user data.

- Platform integration challenges – Working within app restrictions and end-to-end encryption while ensuring user-friendly workflows.

This component of the project addresses these challenges by designing a mobile-sensitive content detection and alert system specifically for WhatsApp and Telegram. The system will scan outgoing media, temporarily block the send button, analyze content, and present a sensitivity alert. This design mimics parental control behavior while respecting user privacy.

## 2. Research Gap

While current research in sensitive content detection has shown strong advances in multimedia classification, there remain significant gaps that limit its application in encrypted messaging platforms such as WhatsApp and Telegram.

Research [1] Lightweight MobileNet-based models for multimedia analysis - proposed MobileNet-based CNN models for multimedia analysis but did not support real-time scanning in messaging workflows or provide alert features.

Research [2] PP-PicoDet: A Better Real-Time Object Detector on Mobile Devices -achieved efficient real-time object detection on mobile devices but not specialized for sensitive content classification or integration into messaging apps.

Research [3] Should I See or Should I Go: Automatic Detection of Sensitive Media in Messaging Apps - explored nudity/violence detection in messaging contexts but lacked lightweight mobile optimization and privacy-preserving, on-device execution.

Thus, there is a clear gap: no existing solution provides a lightweight, on-device, ML-based sensitive content detection and alert system that is optimized for encrypted messaging apps like WhatsApp and Telegram while maintaining privacy, real-time performance, and community-driven extensibility.

| Research Paper | Lightweight ML for Mobile | Real-time Detection on Encrypted Messaging Apps | Alert Mechanism | Privacy-aware On-Device | Community - Driven / Open Source |
|---|---|---|---|---|---|
| Research [1] | Not Implemented | Not Implemented | Not Implemented | Not Implemented | Not Implemented |
| Research [2] | Not Implemented | Not Implemented | Not Implemented | Not Implemented | Not Implemented |
| Research [3] | Not Implemented | Implemented | Not Implemented | Not Implemented | Not Implemented |
| Proposed Solution | Implemented | Implemented | Implemented | Implemented | Implemented |

*Table 1 – Research Gap in Sensitive Content Detection*

## 3. Research Problem

Current sensitive content detection mechanisms in messaging platforms are largely dependent on traditional filtering or cloud-based analysis methods. These approaches are often computationally heavy, require continuous internet access, and are not suitable for real-time integration with encrypted applications like WhatsApp and Telegram.

Another important challenge is that most existing parental control and content moderation tools focus only on static image filtering, leaving video content largely unaddressed. Furthermore, they often lack a user-friendly workflow, such as scanning media before sending, and provide limited or no feedback to the user regarding the sensitivity level of the content.

This situation increases the demand for lightweight, on-device, machine learning–based content detection mechanisms that can work seamlessly within resource-constrained mobile environments. Such solutions must ensure real-time detection and alerting while preserving user privacy and supporting multiple media types.

One promising direction is the use of mobile-optimized deep learning models (e.g., MobileNet, EfficientNet) that can classify sensitive images and videos accurately using smaller computational resources. By integrating these models into messaging workflows, users can be warned or prevented from sharing sensitive content without compromising usability or security.

# II. OBJECTIVES

## 1. Main Objective

The main aim of this research is to design and develop a **lightweight, machine learning–based sensitive content detection and alert system** for WhatsApp and Telegram. The system will scan outgoing images and videos in real time, temporarily disable the send function, analyze the content, and provide a risk-level alert to the user, thereby reducing accidental or harmful sharing of explicit media and improving online safety, especially for minors.

## 2. Sub Objectives

This will be achieved by:

1. Studying existing content detection approaches and identifying their limitations in encrypted messaging platforms, particularly with respect to mobile optimization, video analysis, and privacy concerns.

2. Collecting and curating a dataset of sensitive and non-sensitive images and videos (nudity, violence, explicit content) for training and testing machine learning models.

3. Developing and training lightweight ML models (e.g., MobileNet, EfficientNet) capable of classifying sensitive content efficiently on mobile devices.

4. Designing an in-app scanning workflow that integrates with WhatsApp and Telegram to scan media before sending, with automatic analysis for images and playback analysis for videos.

5. Implementing a risk-based alert mechanism that provides tiered feedback (warning, soft block, allow) based on model confidence levels, ensuring a parental control–like user experience.

6. Optimizing the system for mobile deployment, ensuring minimal resource usage while maintaining high accuracy and user privacy through on-device processing.

7. Testing and validating the system in different scenarios to evaluate performance, accuracy, speed, and usability, ensuring effectiveness in real-world environments.

# III. METHODOLOGY

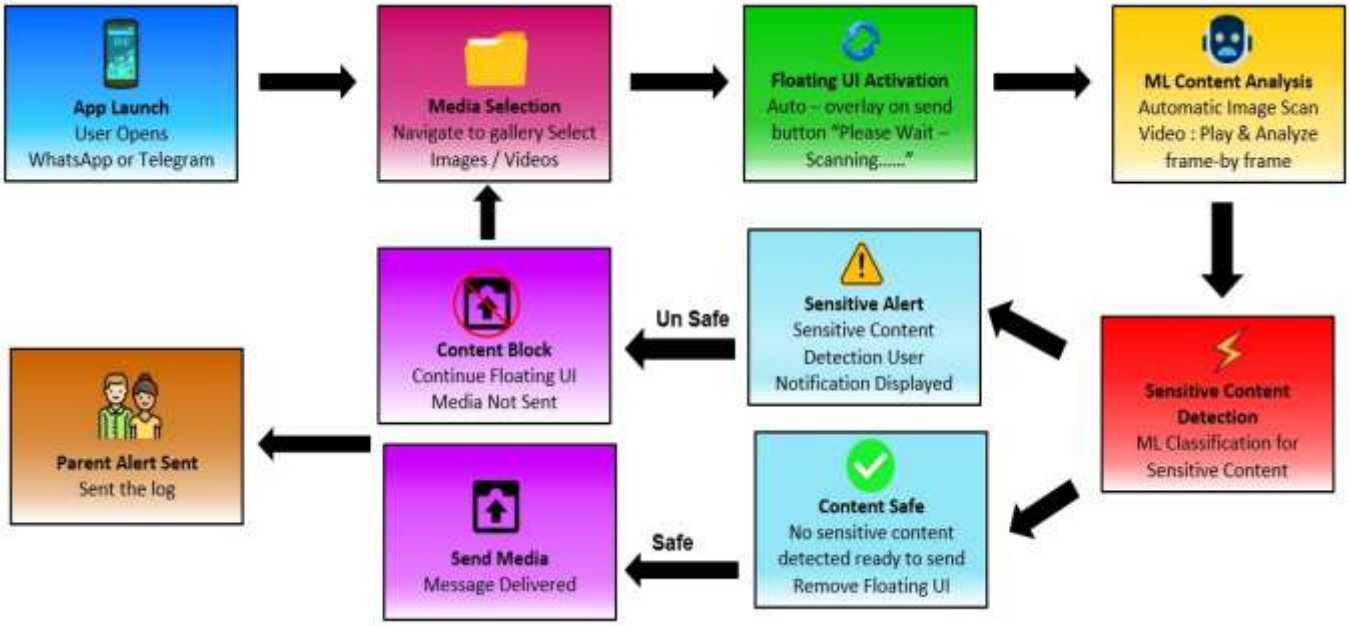


*Figure 1- System Diagram*

## 1. Requirement Gathering

In the requirement gathering process, a combination of literature review, consultation with professionals, and analysis of technical and user requirements was carried out. This process ensured that the proposed solution is feasible, efficient, and addresses real-world challenges in messaging platforms like WhatsApp and Telegram.

- Extensive Literature Review: A review of the latest research and journal articles on machine learning–based image and video classification, parental control systems, and content filtering in encrypted messaging platforms was conducted. This provided insights into existing approaches, their limitations (such as lack of mobile optimization and privacy issues), and potential areas for improvement such as pre-send scanning and tiered alerts.

- Stakeholder Consultation: Discussions were carried out with cybersecurity experts, mobile app developers, and child online safety professionals. These consultations helped to validate the need for real-time sensitive content detection, parental control–like mechanisms, and privacy-preserving solutions. User surveys and feedback sessions further highlighted the importance of integrating a non-intrusive and user-friendly alert system.

- System Requirements Analysis: Technical requirements were identified, including lightweight ML model integration, support for both images and videos, pre-send scanning workflows, and risk-level classification. Performance requirements such as low latency, high accuracy, and efficient mobile resource utilization were also noted. Security and privacy requirements emphasized on-device processing to avoid storing or transmitting sensitive user data.

- Use Case Scenarios: Multiple use case models were developed to demonstrate how the system operates in real-world settings. For instance:

    - A child attempting to share an explicit image via WhatsApp, where the send button is temporarily blocked until a sensitivity alert is shown.

    - A video containing violent scenes being scanned during playback before transmission on Telegram.

    - A user receiving tiered alerts (warning, soft-block, allow) based on model confidence levels.

These scenarios ensure that the proposed solution addresses both functional and practical aspects of sensitive content detection in messaging environments.

## 2. Past Research Analysis

The analysis of prior research provides valuable insights into existing solutions for sensitive content detection and highlights clear gaps that limit safe, private, real-time deployment on encrypted messaging platforms (WhatsApp, Telegram). Below is a concise synthesis of three representative studies and what they contribute - and omit - relative to the goals of a lightweight, on-device, privacy-aware detection & alert system.

### A. Literature Review of Key Studies

Research [1] – MobileNet-based multimedia analysis (2023):
This study proposed MobileNet-based CNN models for multimedia content analysis. While it achieved good accuracy for classifying images, the approach did not support real-time scanning within messaging workflows. In addition, it lacked alert mechanisms to notify users before sharing sensitive media.

Research [2] – PP-PicoDet: Real-Time Object Detection on Mobile (2021):
This work introduced an efficient object detection method optimized for mobile devices. Although it demonstrated strong real-time performance, it was focused on general object recognition rather than detecting sensitive categories like nudity, NIC, or credit card details. Furthermore, it was not designed for integration with messaging applications.

Research [3] – Sensitive Media Detection in Messaging Apps (2021):
This research explored automatic detection of nudity and violence in messaging environments. While it showed the potential of using ML for sensitive media moderation, the models were not lightweight and relied heavily on cloud processing. The solution also lacked privacy-preserving on-device execution and did not include tiered alert mechanisms to guide user decisions.

### B. Identification of Gaps

The reviewed studies highlight several gaps that limit their use in real-world messaging platforms. First, most solutions do not employ lightweight ML models, making them unsuitable for resource-constrained mobile devices. Second, none of the approaches integrate real-time detection workflows into encrypted messaging apps such as WhatsApp or Telegram. Third, there is a lack of tiered alert mechanisms that provide warnings or soft blocks before sending sensitive media. Additionally, existing solutions often depend on cloud-based processing, raising privacy concerns, and very few offer community-driven or open-source frameworks for extensibility.

The proposed solution addresses these gaps by developing a lightweight, on-device ML system capable of detecting nudity, NIC, and credit card details in real time. It integrates with messaging workflows, provides tiered alerts, ensures full privacy through on-device processing, and promotes openness through community-driven improvements.

## 3. Feasibility Study

The technical feasibility of the proposed research incorporates the assessment of whether the tools, technologies, and techniques required in the development and implementation of the ML-based sensitive content detection and alert system are appropriate. This shall be addressed in the following areas:

### A. Tools and Technology

- TensorFlow Lite / PyTorch Mobile: TensorFlow Lite and PyTorch Mobile are optimized deep learning frameworks that enable the deployment of machine learning models directly on mobile devices. These frameworks will be employed to implement lightweight CNNs and transformer-based models for detecting sensitive content in images and videos while ensuring efficiency on resource-constrained smartphones.

- Android Studio & Kotlin/Java: Android Studio will serve as the primary development environment for building the mobile application interface and integrating the detection engine into WhatsApp and Telegram workflows. Kotlin/Java will be used to implement core mobile functionalities, such as scanning triggers, UI for alerts, and integration with pre-send workflows.

- Python: Python will be used for model training, dataset preprocessing, and performance evaluation. Libraries such as OpenCV, TensorFlow, and Scikit-learn will assist in developing and validating the ML models before deployment to mobile platforms.

- SQLite / Local Storage: A lightweight local database (SQLite) or secure storage mechanisms will be used to store sensitivity logs, user settings, and parental control preferences while ensuring data privacy and no leakage outside the device.

- GitHub (Open-Source Framework): GitHub will act as the collaborative open-source platform for sharing code, models, and updates with the research community, supporting community-driven improvement and transparency.

### B. Algorithm and Architectures

- Convolutional Neural Networks (CNNs): CNN-based architectures (e.g., MobileNetV3, EfficientNet-Lite) will be employed for detecting nudity, violence, and explicit content in images due to their efficiency in extracting spatial patterns and suitability for mobile devices. [4]

- Temporal Deep Learning Models: For videos, temporal CNNs or lightweight transformer-based models (e.g., MobileViT, Temporal CNN, or Distilled Vision Transformers) will be used to detect sensitive sequences by analyzing multiple frames and aggregating predictions. [5]

- Transfer Learning & Model Optimization: Pre-trained models will be fine-tuned using curated datasets of sensitive and non-sensitive media. Techniques such as pruning, quantization, and knowledge distillation will be used to reduce model size and improve inference speed on mobile devices. [6]

- Risk Scoring Mechanism: The system will assign a sensitivity alert to each content. Based on this alert, the system will determine whether to show a warning, issue a soft block, or allow the user to proceed. [7]

- Alert Mechanism Architecture: The alert system will be integrated into the app interface, temporarily disabling the "send" button until scanning completes and providing tiered alerts (warning, soft block, override options).

### C. Techniques

- On-device Inference: The ML models will run fully on the device to ensure privacy-preserving detection without exposing user media to cloud servers, aligning with encrypted messaging principles.

- Performance Evaluation: The system will be evaluated on accuracy, false positive/negative rates, inference speed, and resource usage (CPU, memory, battery) to ensure acceptable tradeoffs for real-time usage. [8]

- Dataset Preprocessing & Augmentation: Sensitive and non-sensitive media datasets will undergo preprocessing (resizing, normalization) and augmentation (rotation, noise addition, blurring) to improve model generalization.

- Real-time Video Frame Sampling: Instead of scanning all video frames, key frames will be sampled at intervals and analyzed to reduce latency while maintaining reliable detection accuracy. [9]

- User Interaction Testing: Usability testing will measure user response to tiered alerts, system transparency, and the acceptability of short delays during scanning.

- Privacy & Ethical Handling: Sensitive media will never be stored or uploaded to external servers. Ethical dataset usage and community-driven review mechanisms will ensure fairness and accountability in model training and deployment. [10]

# 4. Project Requirements

## A. Functional Requirements

- Sensitive Content Detection (Images & Videos): The system must automatically scan all selected images and videos before they are sent via WhatsApp or Telegram. Sensitive content such as nudity, violence, or explicit media should be flagged using lightweight ML models optimized for mobile deployment.

- Real-time Alert and Blocking: The system should temporarily disable the "send" button until the scan is completed. Based on the sensitivity alert, the system must generate tiered alerts such as warnings, soft blocks (requiring confirmation), or hard blocks (preventing sending altogether).

- Risk Scoring Mechanism: Each piece of media must be evaluated and assigned a sensitivity alert. The risk alert determines the alert level presented to the user, ensuring flexibility and adaptability of the system in different contexts.

- Privacy Preservation: The detection process must occur entirely on the user's device without uploading media to external servers, maintaining compliance with end-to-end encryption principles of WhatsApp and Telegram.

- Parental Control Features: The system should allow guardians to set thresholds for sensitive detection and enforce stricter blocking rules for minors, ensuring customizable content control.

- Logging & Reporting: The system must provide optional local logging of detection results and blocked media attempts. These logs should remain secure and only accessible to the device owner/guardian.

## B. Non-Functional Requirements

- Accuracy: The ML models should achieve high detection accuracy while minimizing false positives (blocking safe content) and false negatives (missing harmful content). Accuracy must be validated against diverse datasets of sensitive and non-sensitive media.

- Performance: The system must operate in real time, ensuring minimal latency during media scanning. For example, image scanning should complete within seconds, and video scanning should rely on efficient frame sampling to maintain usability.

- Resource Efficiency: The detection system should consume minimal device resources

(CPU, GPU, memory, and battery). Optimized lightweight models (quantization, pruning) should ensure smooth operation on low- and mid-range smartphones.

- Availability & Reliability: The system must remain stable and functional under continuous use, even with multiple media files being scanned simultaneously. It should not crash or significantly slow down messaging applications.

- Security: The system must ensure that sensitive content remains private on the user's device and is not shared externally. Logs, risk alerts, and alerts should be stored securely using encryption, preventing unauthorized access.

- Scalability & Extensibility: The framework should be open-source and designed for extensibility, allowing the community to add new detection categories (e.g., hate symbols, self-harm content) and improve models over time.

- Usability: Alerts and warnings should be clear, user-friendly, and provide simple options (proceed, cancel, override). The system should not overburden users with technical complexity, ensuring smooth integration with everyday messaging use.

## 5. Work Breakdown & Grant Chart

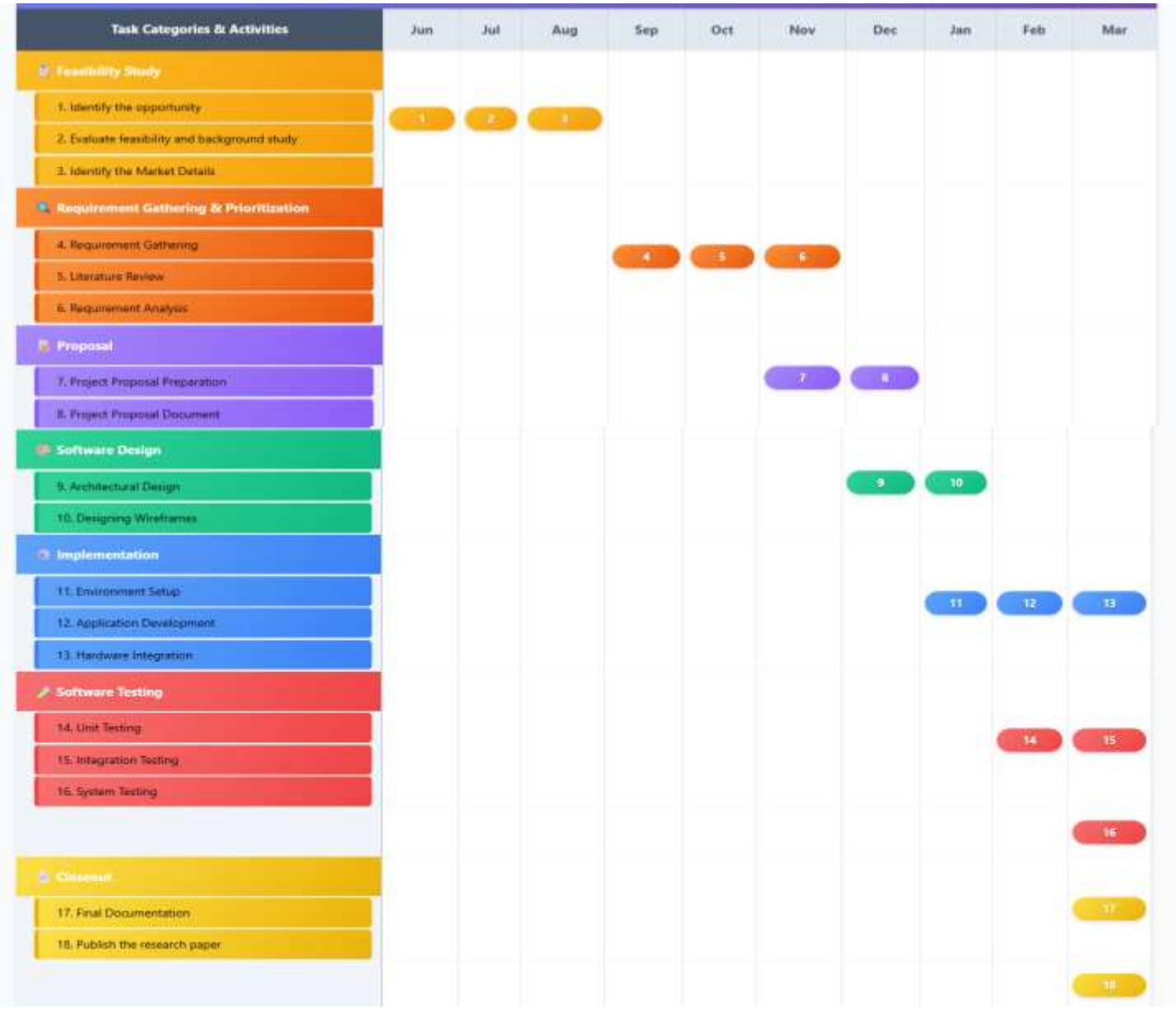


*Figure 2 – Grant chart*

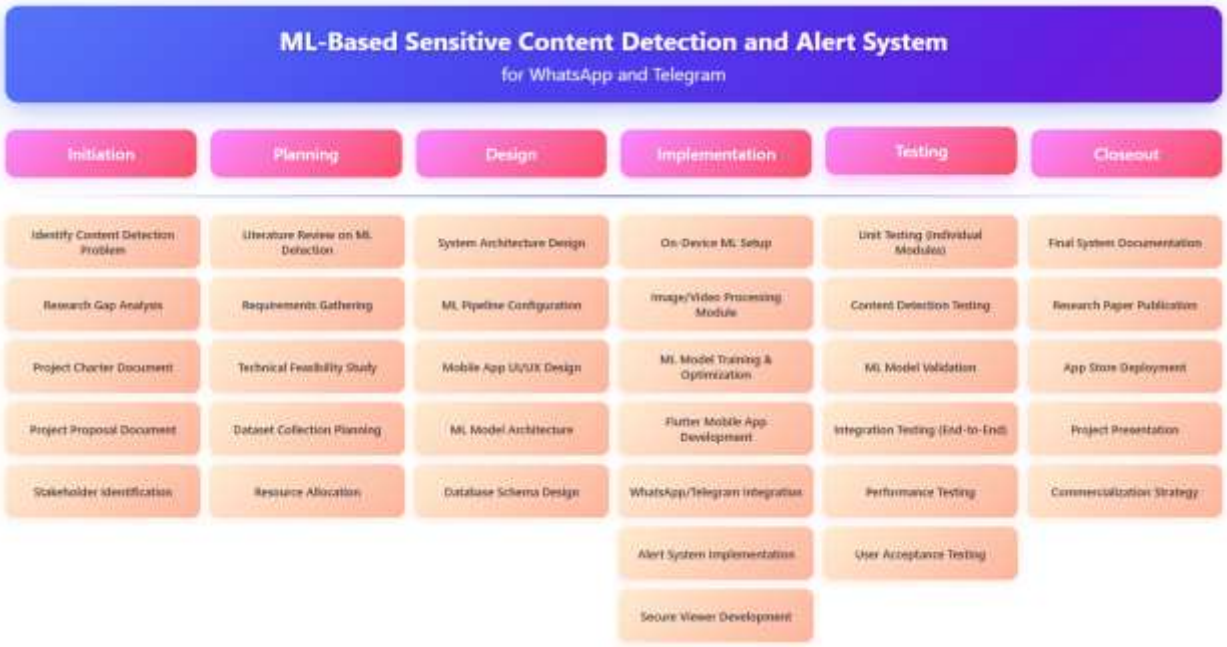


*Figure 3 - Work Breakdown Chart*

## 6. Commercialization

The proposed Sensitive Content Detection System has strong commercialization potential in the growing markets of digital safety and content moderation. With rising risks of sharing nudity, NIC images, and credit card details on messaging platforms, this solution directly addresses a critical demand for real-time, privacy-preserving protection.

The system leverages lightweight ML models optimized for mobile devices, ensuring fast and accurate detection of sensitive images and videos before sending. Its tiered alert mechanism (warnings, soft blocks) and seamless integration with WhatsApp/Telegram make it practical and easy to adopt.

Key Commercialization Points

1. Pricing Packages

- Basic ($15): Nudity, NIC & credit card detection with alerts only.
- Pro ($30): Adds tiered risk scoring (low/medium/high) with both image and video scanning.
- Premium ($50): Full solution with media scanning, tiered alerts, parental/guardian dashboard, and detailed sensitivity reporting.

2. Availability

- Google Play Store (Android)
- Apple App Store (iOS)
- Official Website Download (direct APK/installer option)

3. Promotion Strategy

- Social media campaigns on Facebook, Instagram, TikTok.
- Targeted advertisements for parents, schools, and NGOs.
- Partnerships with child-safety organizations and NGOs.
- Demos and trial versions for schools and institutions to encourage adoption.

4. Unique Selling Proposition (USP)

- Unlike conventional parental apps, this solution provides
- Real-time detection in messaging apps.
- On-device ML models for privacy protection.

- Tiered alert system to guide user choices.
- Lightweight integration without disrupting the user experience.

5. Revenue Model

- Subscription-based packages for individual users and institutions.
- Licensing to messaging apps and smartphone manufacturers.
- Freemium model → free basic detection, with upgrades to Pro/Premium for advanced features.

6. Market Entry Strategy

- Launch through pilot programs in schools, universities, and parent groups.
- Build credibility via collaborations with NGOs and digital-safety advocacy groups.
- Gradual integration with messaging/social media platforms and device vendors.

# IV. BUDGET AND BUDGET JUSTIFICATION

Since the proposed Sensitive Content Detection System is software-oriented, no specialized hardware components are required for its implementation. The primary costs will be associated with cloud computing resources for training lightweight ML models, dataset collection/curation, and dissemination of results through publications. Additional minor costs are anticipated for project execution such as internet access, software licensing, and stationery.

The detailed budget is presented in **Table** below.

| Type | Cost (LKR) |
|---|---|
| Internet and Web Browsing (research, dataset downloads, testing) | 5,000 |
| Cloud Computing / GPU Services (model training, preprocessing) | 20,000 |
| Dataset Acquisition & Annotation (curation, labeling sensitive content) | 12,000 |
| Software Tools & Licenses (Android Studio plugins, ML libraries if premium) | 7,500 |
| Publication Cost (journal/conference submission fees) | 15,000 |
| Stationery & Documentation (reports, printing, presentations) | 5,500 |
| **Total** | **65,000** |

***Table 2 - Budget Analysis***

## Summary

This project proposal aims to enhance digital safety and parental control by developing a lightweight sensitive content detection system for mobile applications. The focus is on addressing the rising concerns of children's exposure to inappropriate images and videos in messaging and social media platforms. The proposed system leverages optimized machine learning models (e.g., MobileNet/EfficientNet) for on-device detection, ensuring efficiency, accuracy, and user privacy. The project will involve a comprehensive literature review, requirement analysis, dataset collection, and ML model development, followed by integration into a mobile environment with a risk scoring and alert mechanism. The solution will be validated through unit, integration, and usability testing to ensure reliability and effectiveness. Additionally, a commercialization strategy will be explored, targeting parents, educational institutions, and digital platforms to provide a scalable and user-friendly tool for safer digital interactions.

# References


[1] A. Sajeeda, P. S. Ansari, and S. R. Naqvi, “Lightweight MobileNet Model for Image Tampering Detection,” International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC), vol. 11, no. 6, pp. 58–64, Jun. 2023.

[2] J. Yu, X. Xu, S. Xu, R. Wang, C. Chen, Y. Wang, and D. Liao, “PP-PicoDet: A Better Real-Time Object Detector on Mobile Devices,” arXiv preprint arXiv:2111.00902, Nov. 2021.

[3] J. Serra, R. Peláez, D. Gómez, and J. Fornés, “Should I See or Should I Go: Automatic Detection of Sensitive Media in Messaging Apps,” in Proceedings of the 27th Brazilian Symposium on Multimedia and the Web (WebMedia ’21), Nov. 2021, pp.

[4] "State-of-the-Art in Nudity Classification: A Comparative Analysis".

[5] "MobileViT," [Online]. Available: https://kornia.readthedocs.io/en/latest/models/vit_mobile.html.

[6] H. L. L. M. Zhuo Li 1, "Model Compression for Deep Neural Networks: A Survey".

[7] "Risk-Based Alerting," [Online]. Available: https://www.deepwatch.com/glossary/risk-based-alerting-rba/.

[8] 2. a. G. V. Konstantina Papafragkaki1, "InferONNX: Practical and Privacy-preserving Machine Learning Inference using Trusted Execution Environments".

[9] Y. Z. Z. T. L. W. G. Wu, "An Explainable Sampling Strategy for Video Action Recognition".

[10] R. S. S. M. P. S. Akanksha Atrey, "Protecting Proprietary Information in On-Device Machine Learning Models".